

Specification for 3.97 inch EPD**Model NO. : DEPG0397BBS77AF11HP****DKE's Confirmation:**

Prepared by	Checked by	Approved by

Customer approval:

Customer	Approved by	Date

Revision History

Version	Content	Date	Producer
2.0	New release	2025/4/11	
2.1	Update: Typical Application Circuit with SPI Interface	2025/9/1	
2.2	Update: 6. Electrical Characteristics	2026/01/13	

CONTENTS

1. Over View	4
2. Features	4
3. Mechanical Specification	5
4. Mechanical Drawing of EPD Module	6
5. Input/output Pin Assignment	7
6. Electrical Characteristics	9
6.1 Absolute Maximum Rating	9
6.2 Panel DC Characteristics	9
6.3 Panel AC Characteristics(Driver IC Internal Regulators)	10
6.4 MCU Interface	11
7. Command Table	16
8. Block Diagram	29
9. Typical Application Circuit with SPI Interface	30
10 Typical Operating Sequence	31
10.1 OTP Operation Flow	31
10.2 OTP Operation Reference Program Code	32
10.3 OTP Partial update Operation Flow	33
11. Reliability Test	34
12. Quality Assurance	35
12.1 Environment	35
12.2 Illuminance	35
12.3 Inspect method	35
12.4 Display area	35
12.5 Ghosting test method	35
12.6 Inspection standard	36
13. Packaging	39
14. Handling, Safety, and Environment Requirements	40

1. Over View

DEPG0397BBS77AF11HP is the display which use the flexible substrate as base plate, with interface and a reference system design. The display is capable to display images at 2-bit white, black full display capabilities. The 3.97 inch active area contains 480(H)×800(V) pixels. The module is a TFT-array driving electrophoresis display, with integrated circuits including gate driver, source driver, MCU interface, timing controller, oscillator, DC-DC, SRAM, LUT, VCOM. Module can be used in portable electronic devices, such as Electronic Shelf Label (ESL) System.

2. Features

- ◆ 480×800 pixels display
- ◆ High contrast High reflectance
- ◆ Ultra wide viewing angle Ultra low power consumption
- ◆ Pure reflective mode
- ◆ Bi-stable display
- ◆ Commercial temperature range
- ◆ Landscape portrait modes
- ◆ Hard-coat antiglare display surface
- ◆ Ultra Low current deep sleep mode
- ◆ On chip display RAM
- ◆ Waveform can stored in On-chip OTP or written by MCU
- ◆ Serial peripheral interface available
- ◆ On-chip oscillator
- ◆ On-chip booster and regulator control for generating VCOM, Gate and Source driving voltage
- ◆ Support partial update mode
- ◆ Built-in temperature sensor

3. Mechanical Specification

Parameter	Specifications	Unit	Remark
Screen Size	3.97	Inch	
Display Resolution	480(H)×800(V)	Pixel	DPI:235
Active Area	51.84×86.40	mm	
Pixel Pitch	0.108×0.108	mm	
Pixel Configuration	Square		
Outline Dimension	56.24(H)×96.62 (V) ×1.0(D)	mm	
Weight	TBD	g	

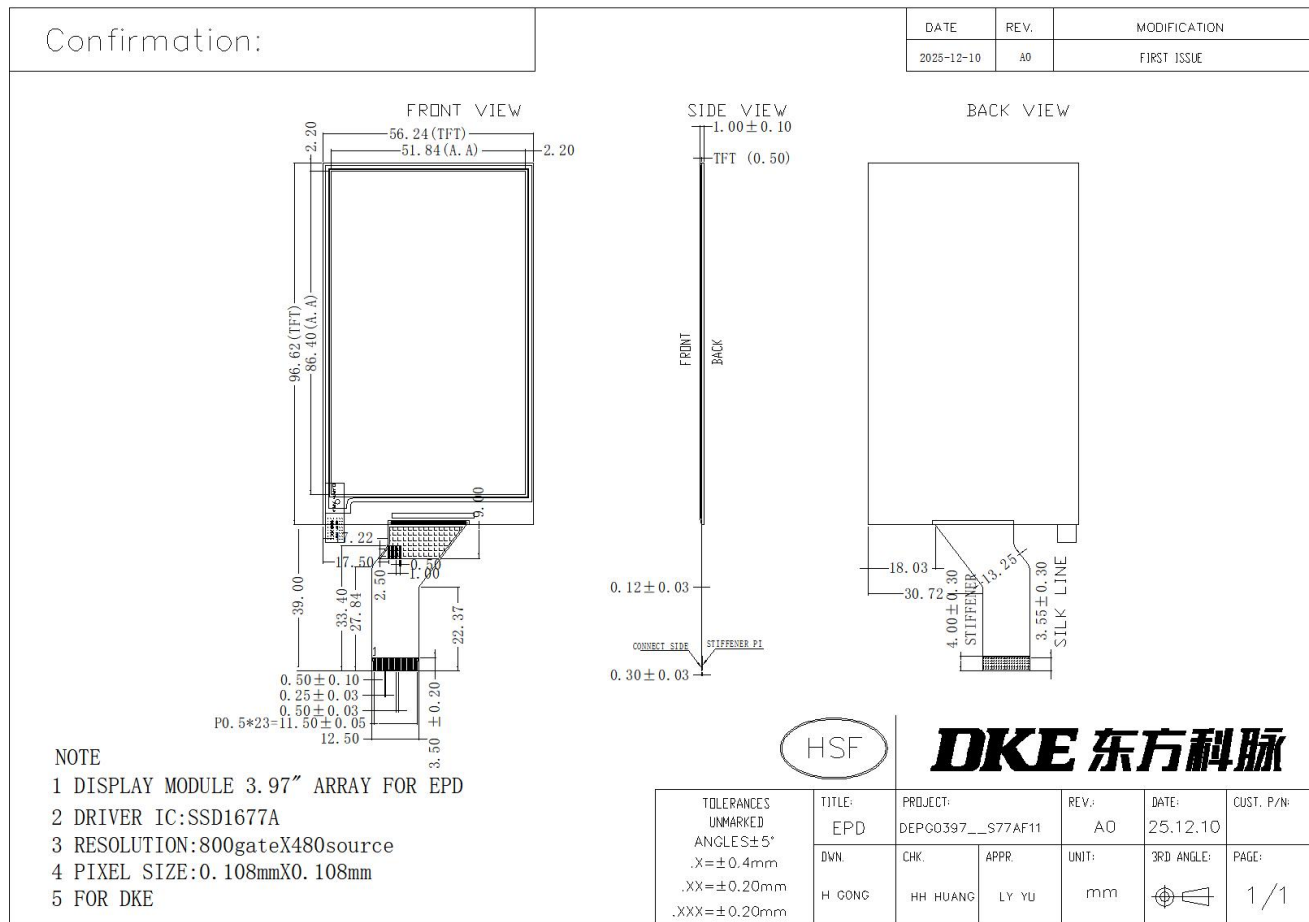
Symbol	Parameter	Conditions	Min	Typ.	Max	Units	Notes
KS	Black State L* value		-	24	30		3-1
WS	White State L* value		58	62	-		3-1
	White Ghosting ΔL	Full Display Mode	-	1	1		3-1
		Partil Display Mode	-	1	-		3-1
		Full-Partil Display Mode	-	1	-		3-1
R	White Reflectivity	White	-	30	-	%	3-1
CR	Contrast Ratio	Indoor	-	7	-		3-1 3-2
GN	2Grey Level	-	-	-	-		
Life		Temp:23±3°C Humidity:55±10%RH		5years			3-3

Notes: 3-1. Luminance meter: Eye-One Pro Spectrophotometer.

3-2. CR=Surface Reflectance with all white pixel/Surface Reflectance with all black pixels.

3-3. When the product is stored. The display screen should be kept white and face up.

4.Mechanical Drawing of EPD Module



5. Input/output Pin Assignment

No.	Name	I/O	Description	Remark
1	PAD A	NC	Connect to Test A on FPC	
2	GDR	O	N-Channel MOSFET Gate Drive Control	
3	RESE	I	Current Sense Input for the Control Loop	
4	PAD B	NC	Connect to Test B on FPC	
5	VDHR	C	Positive Source driving voltage 2	
6	PAD C	NC	Connect to Test C on FPC	
7	PAD D	NC	Connect to Test D on FPC	
8	BS	I	Bus Interface selection pin	Note 5-5
9	BUSY_N	O	Busy state output pin	Note 5-4
10	RST_N	I	Reset signal input. Active Low.	Note 5-3
11	DC	I	Data /Command control pin	Note 5-2
12	CSB	I	Chip select input pin	Note 5-1
13	SCL	I	Serial Clock pin (SPI)	
14	SDA	I/O	Serial Data pin (SPI)	
15	VDDIO	P	Power Supply for interface logic pins It should be connected with VCI	
16	VDD	P	Power Supply for the chip	
17	GND	P	Ground	
18	VDDD	C	Core logic power pin VDD can be regulated internally from VCI. A capacitor should be connected between VDD and VSS	
19	VPP	P	FOR TEST	Keep Open
20	VSH	C	Positive Source driving voltage	
21	VGH	C	Power Supply pin for Positive Gate driving voltage and VSH1	
22	VSL	C	Negative Source driving voltage	
23	VGL	C	Power Supply pin for Negative Gate driving voltage VCOM and VSL	
24	VCOM	C	VCOM driving voltage	

I = Input Pin, O =Output Pin, I/O = Bi-directional Pin (Input/output), P = Power Pin, C = Capacitor Pin

Note 5-1: This pin (CS#) is the chip select input connecting to the MCU. The chip is enabled for MCU communication only when CS# is pulled LOW.

Note 5-2: This pin is (D/C#) Data/Command control pin connecting to the MCU in 4-wire SPI mode. When the pin is pulled HIGH, the data at SDA will be interpreted as data. When the pin is pulled LOW, the data at SDA will be interpreted as command.

Note 5-3: This pin (RES#) is reset signal input. The Reset is active low.

Note 5-4: This pin is Busy state output pin. When Busy is High, the operation of chip should not be interrupted, command should not be sent. The chip would put Busy pin High when -Outputting display waveform -Communicating with digital temperature sensor

Note 5-5: Bus interface selection pin

BS1 State	MCU Interface
L	4-lines serial peripheral interface(SPI) - 8 bits SPI
H	3- lines serial peripheral interface(SPI) - 9 bits SPI

6. Electrical Characteristics

6.1 Absolute Maximum Rating

Parameter	Symbol	Rating	Unit
Logic supply voltage	V _{CI} , V _{DDIO}	-0.5 to +4.0	V
Logic Input voltage	V _{IN}	-0.5 to V _{DDIO} +0.5	V
Logic Output voltage	V _{OUT}	-0.5 to V _{DDIO} +0.5	V
Operating Temp range	TOPR	0 to +50	°C.
Storage Temp range	TSTG	-25 to +70	°C.
Optimal Storage Temp	TSTGo	23±3	°C.
Optimal Storage Humidity	HSTGo	55±10	RH

Note:

1. Maximum ratings are those values beyond which damages to the device may occur. Functional operation should be restricted to the limits in the Panel DC Characteristics tables.
2. The display screen should be kept white and face up during storage. Please refer to the [Reliability Test] section.

6.2 Panel DC Characteristics

The following specifications apply for: V_{SS}=0V, V_{CI}=3.0V, TOPR = 25°C.

Parameter	Symbol	Condition	Applicable pin	Min.	Typ.	Max.	Unit
Single ground	V _{SS}	-		-	0	-	V
Logic supply voltage	V _{CI}	-	V _{CI}	2.2	3.0	3.3	V
Power for interface logic pins	V _{DDIO}	-	V _{DDIO}	2.2	-	3.3	V
Core logic voltage	V _{DD}		V _{DD}	1.7	1.8	1.9	V
VCOM_DC output voltage	V _{com}	-	VCOM	-4.0	-	-0.1	V
High level input voltage	V _{IH}	-	-	0.8 V _{DDIO}	-	-	V
Low level input voltage	V _{IL}	-	-	-	-	0.2 V _{DDIO}	V
High level output voltage	V _{OH}	I _{OH} = -100uA	-	0.9 V _{DDIO}	-	-	V
Low level output voltage	V _{OL}	I _{OL} = 100uA	-	-	-	0.1 V _{DDIO}	V
Typical power	P _{TYP}	V _{CI} = 3.0V	-	-		-	mW
Deep sleep mode	P _{STPY}	V _{CI} = 3.0V	-	-	0.0009	-	mW

Typical operating current	Iopr_VCI	V _{CI} = 3.0V	-	-	5.5	-	mA
Image update time	-	25 °C	-	-	3.2	-	sec
Typical peak current	Iopr_VCI	2.2~3.7v			90	-	mA
Sleep mode current	Islp_VCI	DC/DC off No clock No input load Ram data retain	-	-	25	40	uA
Deep sleep mode current	Idslp_VCI	DC/DC off No clock No input load Ram data not retain	-	-	1	5	uA

Notes: 1. The typical power measurement method is as follows, and the output is reported according to the test screen provided by the DKE.



2. The deep sleep power is the consumed power when the panel controller is in deep sleep mode.
3. The listed electrical characteristics are only guaranteed under the controller & waveform provided by DKE.
4. Electrical measurement: Tektronix oscilloscope - MSO044,
Tektronix current probe - TCPA300.

6.3 Panel AC Characteristics(Driver IC Internal Regulators)

The following specifications apply for: VSS=0V, VCI=3.0V, TOPR =23°C.

Parameter	Symbol	Condition	Applicable pin	Min.	Typ.	Max.	Unit
VCOM output voltage	VCOM	VDD=3V After PON Command VGH=20V VGL=-VGH VSH=15V	VCOM	-2.2	2	-1.8	V
Positive Source output voltage	V _{SH}		S ₀ ~S ₉₅₉	+14.8	+15	+15.2	V
Negative Source output voltage	V _{SL}		S ₀ ~S ₉₅₉	-15.2	-15	-14.8	V

Positive gate output voltage	Vgh	VSH2=15V VSL=-15V VCOM = -2V No waveform transitions. No output load No RAM read/write	G ₀ ~G ₆₃₉	+19.5	+20	+20.5	V
Negative gate output voltage	Vgl		G ₀ ~G ₆₃₉	-20.5	-20	-19.5	V

6.4 MCU Interface

6.4.1 MCU Interface Selection

The pin assignment at different interface mode is summarized in Table 6-4-1. Different MCU mode can be set by hardware selection on BS1 pins. The display panel only supports 4-wire SPI or 3-wire SPI interface mode.

Pin Name	Data/Command Interface		Control Signal		
Bus interface	SDA	SCL	CS#	D/C#	RES#
BS1=L 4-wire SPI	SDA	SCL	CS#	D/C#	RES#
BS1=H 3-wire SPI	SDA	SCL	CS#	L	RES#

6.4.2 MCU Serial Interface (4-wire SPI)

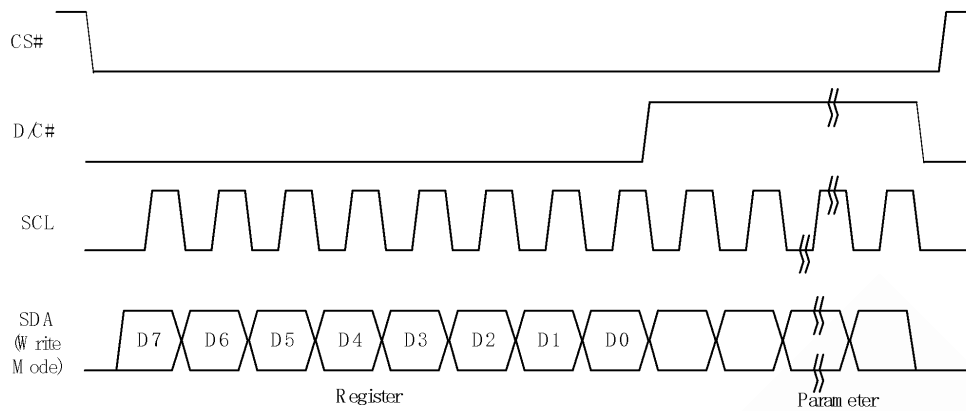
The serial interface consists of serial clock SCL, serial data SDA, D/C#, CS#. This interface supports Write mode and Read mode.

Function	CS#	D/C#	SCL
Write command	L	L	↑
Write data	L	H	↑

Note: ↑ stands for rising edge of signal

In the write mode SDA is shifted into an 8-bit shift register on every rising edge of SCL in the order of D7, D6, ... D0. The level of D/C# should be kept over the whole byte. The data byte in the shift register is written to the Graphic Display Data RAM /Data Byte register or command Byte register according to D/C# pin.

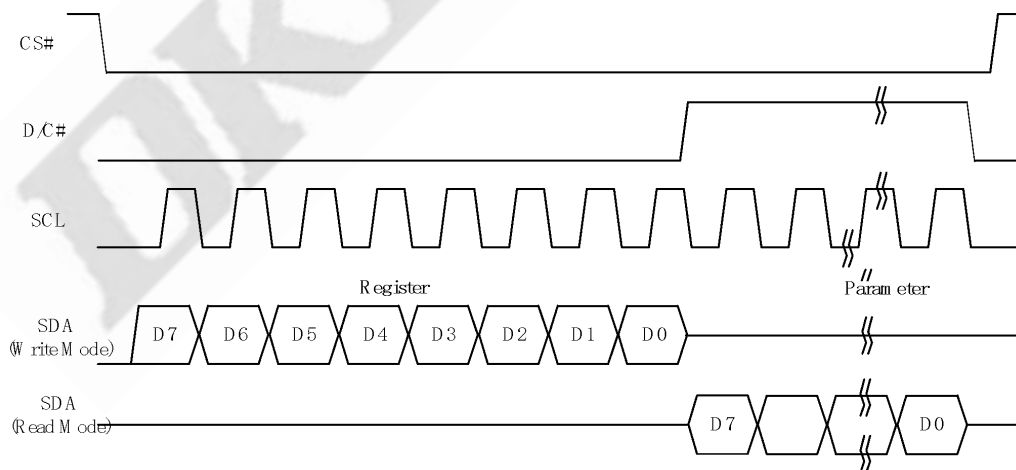
Figure 6-1: Write procedure in 4-wire SPI mode



In the Read mode:

1. After driving CS# to low, MCU need to define the register to be read.
2. SDA is shifted into an 8-bit shift register on every rising edge of SCL in the order of D7, D6, ... D0 with D/C# keep low.
3. After SCL change to low for the last bit of register, D/C# need to drive to high.
4. SDA is shifted out an 8-bit data on every falling edge of SCL in the order of D7, D6, ... D0.
5. Depending on register type, more than 1 byte can be read out. After all byte are read, CS# need to drive to high to stop the read operation.

Figure 6-2: Read procedure in 4-wire SPI mode



6.4.3 MCU Serial Interface (3-wire SPI)

The 3-wire serial interface consists of serial clock SCL, serial data SDA and CS#. This

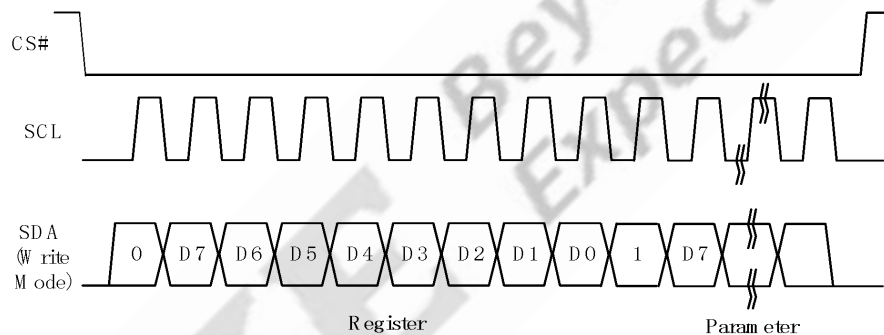
interface also supports Write mode and Read mode.

The operation is similar to 4-wire serial interface while D/C# pin is not used. There are altogether 9-bits will be shifted into the shift register on every ninth clock in sequence: D/C# bit, D7 to D0 bit. The D/C# bit (first bit of the sequential data) will determine the following data byte in the shift register is written to the Display Data RAM (D/C# bit = 1) or the command register (D/C# bit = 0).

Function	CS#	D/C#	SCL
Write command	L	Tie	↑
Write data	L	Tie	↑

Note: ↑ stands for rising edge of signal

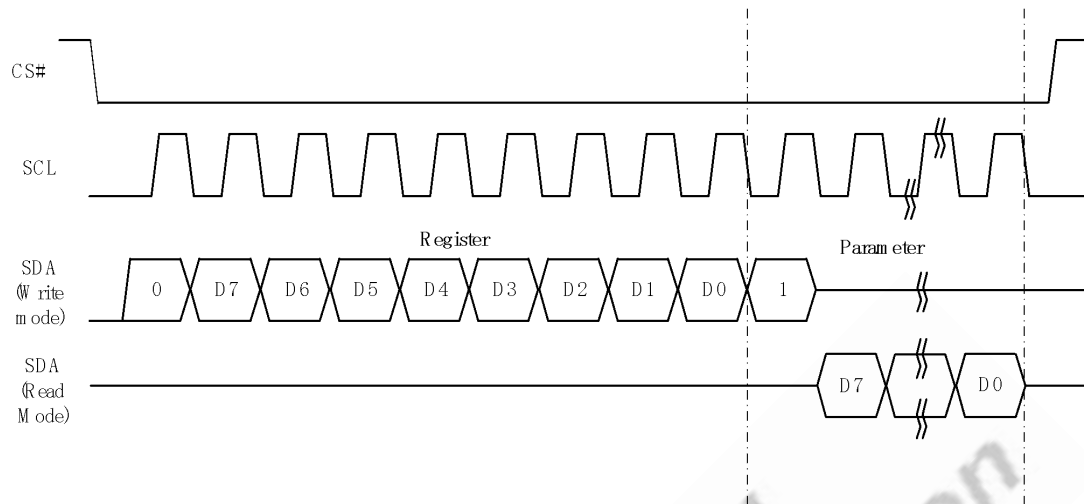
Figure 6-3: Write procedure in 3-wire SPI mode



In the Read mode:

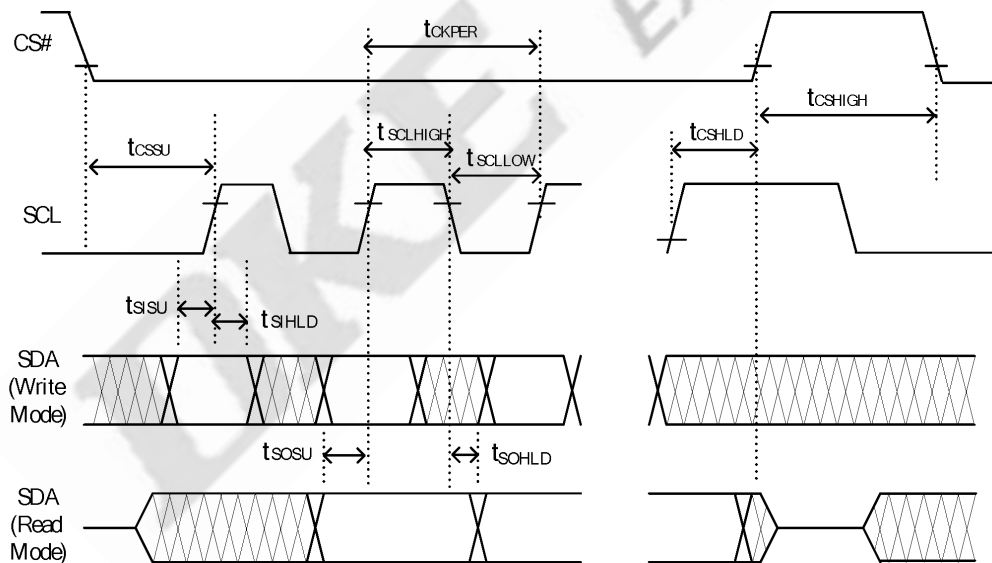
1. After driving CS# to low, MCU need to define the register to be read.
2. D/C=0 is shifted thru SDA with one rising edge of SCL
3. SDA is shifted into an 8-bit shift register on every rising edge of SCL in the order of D7, D6, ... D0.
4. D/C=1 is shifted thru SDA with one rising edge of SCL
5. SDA is shifted out an 8-bit data on every falling edge of SCL in the order of D7, D6, ... D0.
6. Depending on register type, more than 1 byte can be read out. After all byte are read, CS# need to drive to high to stop the read operation.

Figure 6-4: Read procedure in 3-wire SPI mode



6.4.4 Interface Timing

The following specifications apply for: VSS=0V, VCI=3.0V, T_{OPR} =25°C.



Changed Diagram

Serial Interface Timing Characteristics

(VCI - VSS = 2.2V to 3.7V, TOPR = 25°C, CL=20pF)

Write mode

Symbol	Parameter	Min	Typ.	Max	Unit
fSCL	SCL frequency (Write Mode)			20	MHz
tCSSU	Time CS# has to be low before the first rising edge of SCLK	20			ns
tCSHLD	Time CS# has to remain low after the last falling edge of SCLK	20			ns
tCSHIGH	Time CS# has to remain high between two transfers	100			ns
tSCLHIGH	Part of the clock period where SCL has to remain high	25			ns
tSCLLOW	Part of the clock period where SCL has to remain low	25			ns
tSISU	Time SI (SDA Write Mode) has to be stable before the next rising edge of SCL	10			ns
tSIHLD	Time SI (SDA Write Mode) has to remain stable after the rising edge of SCL	40			ns

Read mode

Symbol	Parameter	Min	Typ.	Max	Unit
fSCL	SCL frequency (Read Mode)			2.5	MHz
tCSSU	Time CS# has to be low before the first rising edge of SCLK	100			ns
tCSHLD	Time CS# has to remain low after the last falling edge of SCLK	50			ns
tCSHIGH	Time CS# has to remain high between two transfers	250			ns
tSCLHIGH	Part of the clock period where SCL has to remain high	180			ns
tSCLLOW	Part of the clock period where SCL has to remain low	180			ns
tSOSU	Time SO(SDA Read Mode) will be stable before the next rising edge of SCL		50		ns
tSOHLD	Time SO (SDA Read Mode) will remain stable after the falling edge of SCL		0		ns

7. Command Table

Table 7-1: Command Table

Command Table												Command	Description																																											
R/W#	D/C#	Hex	D7	D6	D5	D4	D3	D2	D1	D0																																														
0	0	01	0	0	0	0	0	0	0	1	Driver Output control	Gate setting A[9:0]= 2A7h [POR], 680 MUX MUX Gate lines setting as (A[9:0] + 1).																																												
0	1		A7	A6	A5	A4	A3	A2	A1	A0																																														
0	1		0	0	0	0	0	0	A0	A8																																														
0	1		0	0	0	0	0	B2	B1	B0																																														
												B[2:0] = 000 [POR]. Gate scanning sequence and direction B[2]: GD Selects the 1st output Gate GD=0 [POR], G0 is the 1st gate output channel, gate output sequence is G0,G1, G2, G3, ... GD=1, G1 is the 1st gate output channel, gate output sequence is G1, G0, G3, G2, ... B[1]: SM Change scanning order of gate driver. SM=0 [POR], G0, G1, G2, G3...679 (left and right gate interlaced) SM=1, G0, G2, G4 ...G678, G1, G3, ...G679 B[0]: TB TB = 0, Scan from G0 to G679 TB = 1, Reserved																																												
0	0	03	0	0	0	0	0	0	1	1	Gate Driving voltage Control	Set Gate driving voltage A[4:0] = 00h [POR] VGH setting from 12V to 20V																																												
0	1		0	0	0	A4	A3	A2	A1	A0																																														
												<table><tr><th>A[4:0]</th><th>VGH</th><th>A[4:0]</th><th>VGH</th></tr><tr><td>00h</td><td>20</td><td>10h</td><td>16.5</td></tr><tr><td>07h</td><td>12</td><td>11h</td><td>17</td></tr><tr><td>08h</td><td>12.5</td><td>12h</td><td>17.5</td></tr><tr><td>09h</td><td>13</td><td>13h</td><td>18</td></tr><tr><td>0Ah</td><td>13.5</td><td>14h</td><td>18.5</td></tr><tr><td>0Bh</td><td>14</td><td>15h</td><td>19</td></tr><tr><td>0Ch</td><td>14.5</td><td>16h</td><td>19.5</td></tr><tr><td>0Dh</td><td>15</td><td>17h</td><td>20</td></tr><tr><td>0Eh</td><td>15.5</td><td>Other</td><td>NA</td></tr><tr><td>0Fh</td><td>16</td><td></td><td></td></tr></table>	A[4:0]	VGH	A[4:0]	VGH	00h	20	10h	16.5	07h	12	11h	17	08h	12.5	12h	17.5	09h	13	13h	18	0Ah	13.5	14h	18.5	0Bh	14	15h	19	0Ch	14.5	16h	19.5	0Dh	15	17h	20	0Eh	15.5	Other	NA	0Fh	16		
A[4:0]	VGH	A[4:0]	VGH																																																					
00h	20	10h	16.5																																																					
07h	12	11h	17																																																					
08h	12.5	12h	17.5																																																					
09h	13	13h	18																																																					
0Ah	13.5	14h	18.5																																																					
0Bh	14	15h	19																																																					
0Ch	14.5	16h	19.5																																																					
0Dh	15	17h	20																																																					
0Eh	15.5	Other	NA																																																					
0Fh	16																																																							

Command Table														
R/W#	D/C#	Hex	D7	D6	D5	D4	D3	D2	D1	D0	Command	Description		
0	0	04	0	0	0	0	0	1	0	0	Source Driving voltage Control	Set Source driving voltage A[7:0] = 41h [POR], VSH1 at 15V B[7:0] = A8h [POR], VSH2 at 5V. C[7:0] = 32h [POR], VSL at -15V		
0	1		A7	A6	A5	A4	A3	A2	A1	A0				
0	1		B7	B6	B5	B4	B3	B2	B1	B0				
0	1		C7	C6	C5	C4	C3	C2	C1	C0				
B[7] = 1, VSH2 voltage setting from 2.4V to 8.8V														
A[7]/B[7] = 0, VSH1/VSH2 voltage setting from 9V to 17V														
C[7] = 0, VSL setting from -9V to -17V														
A/B[7:0]			VSH1/VSH2		A/B[7:0]			VSH1/VSH2		C[7:0]			VSL	
8Eh			2.4		AFh			5.7		23h			9	
8Fh			2.5		B0h			5.8		24h			9.2	
90h			2.6		B1h			5.9		25h			9.4	
91h			2.7		B2h			6		26h			9.6	
92h			2.8		B3h			6.1		27h			9.8	
93h			2.9		B4h			6.2		28h			10	
94h			3		B5h			6.3		29h			10.2	
95h			3.1		B6h			6.4		2Ah			10.4	
96h			3.2		B7h			6.5		2Bh			10.6	
97h			3.3		B8h			6.6		2Ch			10.8	
98h			3.4		B9h			6.7		2Dh			11	
99h			3.5		BAh			6.8		2Eh			11.2	
9Ah			3.6		BBh			6.9		2Fh			11.4	
9Bh			3.7		BCh			7		30h			11.6	
9Ch			3.8		BDh			7.1		31h			11.8	
9Dh			3.9		BEh			7.2		32h			12	
9Eh			4		BFh			7.3		33h			12.2	
9Fh			4.1		Ch			7.4		34h			12.4	
A0h			4.2		C1h			7.5		35h			12.6	
A1h			4.3		C2h			7.6		36h			12.8	
A2h			4.4		C3h			7.7		37h			13	
A3h			4.5		C4h			7.8		38h			13.2	
A4h			4.6		C5h			7.9		39h			13.4	
A5h			4.7		C6h			8		3Ah			13.6	
A6h			4.8		C7h			8.1		3Bh			13.8	
A7h			4.9		C8h			8.2						
A8h			5		C9h			8.3						
A9h			5.1		CAh			8.4						
AAh			5.2		CBh			8.5						
ABh			5.3		Ch			8.6						
ACh			5.4		CDh			8.7						
ADh			5.5		CEh			8.8						
AEh			5.6		Other			NA						

Remark:
VSH1> VSH2

Remark:
VSH1> VSH2

R/W#	D/C#	Hex	D7	D6	D5	D4	D3	D2	D1	D0	Command	Description																								
0	0	08	0	0	0	0	1	0	0	0	Initial Code Setting OTP Program	Program Initial Code Setting The command required CLKEN=1. Refer to Register 0x22 for detail. BUSY pad will output high during operation.																								
0	0	09	0	0	0	0	1	0	0	1	Write Register for Initial Code Setting	Write Register for Initial Code Setting Selection A[7:0] ~ D[7:0]: Reserved Details refer to Application Notes of Initial Code Setting																								
0	1		A ₇	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀																										
0	1		B ₇	B ₆	B ₅	B ₄	B ₃	B ₂	B ₁	B ₀																										
0	1		C ₇	C ₆	C ₅	C ₄	C ₃	C ₂	C ₁	C ₀																										
0	1		D ₇	D ₆	D ₅	D ₄	D ₃	D ₂	D ₁	D ₀																										
0	0	0A	0	0	0	0	1	0	1	0	Read Register for Initial Code Setting	Read Register for Initial Code Setting																								
0	0	0C	0	0	0	0	1	1	0	0	Booster Soft-start Control	This command is used to control the inrush current for the booster. Two level of strengths can be selected for the booster																								
0	1		A ₇	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀																										
0	1		B ₇	B ₆	B ₅	B ₄	B ₃	B ₂	B ₁	B ₀																										
0	1		C ₇	C ₆	C ₅	C ₄	C ₃	C ₂	C ₁	C ₀																										
0	1		D ₇	D ₆	D ₅	D ₄	D ₃	D ₂	D ₁	D ₀																										
			E ₇	E ₆	E ₅	E ₄	E ₃	E ₂	E ₁	E ₀		<table><tr><td></td><td colspan="5">Data bytes</td></tr><tr><td></td><td>A[7:0]</td><td>B[7:0]</td><td>C[7:0]</td><td>D[7:0]</td><td>E[7:0]</td></tr><tr><td>Level 1</td><td>AE</td><td>C7</td><td>C3</td><td>C0</td><td>40</td></tr><tr><td>Level 2</td><td>AE</td><td>C7</td><td>C3</td><td>C0</td><td>80</td></tr></table>		Data bytes						A[7:0]	B[7:0]	C[7:0]	D[7:0]	E[7:0]	Level 1	AE	C7	C3	C0	40	Level 2	AE	C7	C3	C0	80
	Data bytes																																			
	A[7:0]	B[7:0]	C[7:0]	D[7:0]	E[7:0]																															
Level 1	AE	C7	C3	C0	40																															
Level 2	AE	C7	C3	C0	80																															
0	0	10	0	0	0	1	0	0	0	0	Deep Sleep mode	Deep Sleep mode Control: <table><tr><td>A[1:0]</td><td>Description</td></tr><tr><td>00</td><td>Normal Mode [POR]</td></tr><tr><td>11</td><td>Enter Deep Sleep Mode</td></tr></table> After this command initiated, the chip will enter Deep Sleep Mode, BUSY pad will keep output high. Remark: To Exit Deep Sleep mode, User required to send HWRESET to the driver	A[1:0]	Description	00	Normal Mode [POR]	11	Enter Deep Sleep Mode																		
A[1:0]	Description																																			
00	Normal Mode [POR]																																			
11	Enter Deep Sleep Mode																																			
0	1		0	0	0	0	0	0	A ₁	A ₀																										

Command Table												Command	Description
R/W#	D/C#	Hex	D7	D6	D5	D4	D3	D2	D1	D0			
0	0	11	0	0	0	1	0	0	0	1		Data Entry mode setting	Define data entry sequence A[2:0] = 011 [POR] A [1:0] = ID[1:0] Address automatic increment / decrement setting The setting of incrementing or decrementing of the address counter can be made independently in each upper and lower bit of the address. 00 –Y decrement, X decrement, 01 –Y decrement, X increment, 10 –Y increment, X decrement, 11 –Y increment, X increment [POR] A[2] = AM Set the direction in which the address counter is updated automatically after data are written to the RAM. AM= 0, the address counter is updated in the X direction. [POR] AM = 1, the address counter is updated in the Y direction.
0	1		0	0	0	0	0	A ₂	A ₁	A ₀			
0	0	12	0	0	0	1	0	0	1	0		SW RESET	It resets the commands and parameters to their S/W Reset default values except R10h-Deep Sleep Mode During operation, BUSY pad will output high. Note: RAM are unaffected by this command.
0	0	14	0	0	0	1	0	1	0	0		HV Ready Detection	HV ready detection A[6:0] = 00h [POR] The command required CLKEN=1 and ANALOGEN=1. Refer to Register 0x22 for detail. After this command initiated, HV Ready detection starts. BUSY pad will output high during detection. The detection result can be read from the Status Bit Read (Command 0x2F).
0	1		0	A ₆	A ₅	A ₄	0	A ₂	A ₁	A ₀			A[6:4]=n for cool down duration: 10ms x (n+1) A[2:0]=m for number of Cool Down Loop to detect. The max HV ready duration is 10ms x (n+1) x (m) HV ready detection will be trigger after each cool down time. The detection will be completed when HV is ready. For 1 shot HV ready detection, A[7:0] can be set as 00h.

Command Table												Command	Description														
R/W#	DI/CS#	Hex	D7	D6	D5	D4	D3	D2	D1	D0																	
0	0	15	0	0	0	1	0	1	0	1	VCI Detection	VCI Detection A[2:0] = 100 [POR] , Detect level at 2.3V A[2:0] : VCI level Detect	<table><tr><th>A[2:0]</th><th>VCI level</th></tr><tr><td>011</td><td>2.2V</td></tr><tr><td>100</td><td>2.3V</td></tr><tr><td>101</td><td>2.4V</td></tr><tr><td>110</td><td>2.5V</td></tr><tr><td>111</td><td>2.6V</td></tr><tr><td>Other</td><td>NA</td></tr></table>	A[2:0]	VCI level	011	2.2V	100	2.3V	101	2.4V	110	2.5V	111	2.6V	Other	NA
A[2:0]	VCI level																										
011	2.2V																										
100	2.3V																										
101	2.4V																										
110	2.5V																										
111	2.6V																										
Other	NA																										
0	1		0	0	0	0	0	A ₂	A ₁	A ₀																	
The command required CLKEN=1 and ANALOGEN=1 Refer to Register 0x22 for detail.																											
After this command initiated, VCI detection starts. BUSY pad will output high during detection. The detection result can be read from the Status Bit Read (Command 0x2F).																											
0	0	18	0	0	0	1	1	0	0	0	Temperature Sensor Control	Temperature Sensor Selection A[7:0] = 48h [POR], external temperature sensor A[7:0] = 80h Internal temperature sensor															
0	1		A ₇	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀																	
0	0	1A	0	0	0	1	1	0	1	0	Temperature Sensor Control (Write to temperature register)	Write to temperature register. A[11:0] = 7FFh [POR]															
0	1		A ₁₁	A ₁₀	A ₉	A ₈	A ₇	A ₆	A ₅	A ₄																	
0	1		A ₃	A ₂	A ₁	A ₀	0	0	0	0																	
0	0	1B	0	0	0	1	1	0	1	1	Temperature Sensor Control (Read from temperature register)	Read from temperature register.															
1	1		A ₁₁	A ₁₀	A ₉	A ₈	A ₇	A ₆	A ₅	A ₄																	
1	1		A ₃	A ₂	A ₁	A ₀	0	0	0	0																	

Command Table												Command	Description
R/W#	DI/CS#	Hex	D7	D6	D5	D4	D3	D2	D1	D0			
0	0	1C	0	0	0	1	1	1	0	0		Temperature Sensor Control (Write Command to External temperature sensor)	Write Command to External temperature sensor.
0	1		A ₇	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀			A[7:0] = 00h [POR],
0	1		B ₇	B ₆	B ₅	B ₄	B ₃	B ₂	B ₁	B ₀			B[7:0] = 00h [POR],
0	1		C ₇	C ₆	C ₅	C ₄	C ₃	C ₂	C ₁	C ₀			C[7:0] = 00h [POR],
													A[7:6]
													A[7:6] Select no of byte to be sent
													00 Address + pointer
													01 Address + pointer + 1st parameter
													10 Address + pointer + 1st parameter + 2nd pointer
													11 Address
													A[5:0] – Pointer Setting
													B[7:0] – 1 st parameter
													C[7:0] – 2 nd parameter
													The command required CLKEN=1.
													Refer to Register 0x22 for detail.
													After this command initiated, Write Command to external temperature sensor starts. BUSY pad will output high during operation.
0	0	20	0	0	1	0	0	0	0	0		Master Activation	Activate Display Update Sequence
													The Display Update Sequence Option is located at R22h.
													BUSY pad will output high during operation. User should not interrupt this operation to avoid corruption of panel images.
0	0	21	0	0	1	0	0	0	0	1		Display Update Control 1	RAM content option for Display Update
0	1		A ₇	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀			A[7:0] = 00h [POR]
													A[7:4] Red RAM option
													0000 Normal
													0100 Bypass RAM content as 0
													1000 Inverse RAM content
													A[3:0] BW RAM option
													0000 Normal
													0100 Bypass RAM content as 0
													1000 Inverse RAM content

Command Table											Command	Description																										
R/W#	DI/C#	Hex	D7	D6	D5	D4	D3	D2	D1	D0																												
0	0	22	0	0	1	0	0	0	1	0	Display Update Control 2	Display Update Sequence Option: Enable the stage for Master Activation A[7:0]= FFh (POR)																										
0	1		A7	A6	A5	A4	A3	A2	A1	A0																												
											<table><tr><th>Operating sequence</th><th>Parameter (in Hex)</th></tr><tr><td>Enable clock signal</td><td>80</td></tr><tr><td>Disable clock signal</td><td>01</td></tr><tr><td>Enable clock signal → Enable Analog</td><td>C0</td></tr><tr><td>Disable Analog → Disable clock signal</td><td>03</td></tr><tr><td>Enable clock signal → Load LUT with DiSPLAY Mode 1 → Disable clock signal</td><td>91</td></tr><tr><td>Enable clock signal → Load LUT with DiSPLAY Mode 2 → Disable clock signal</td><td>99</td></tr><tr><td>Enable clock signal → Load temperature value from I2C Single Master Interface → Load LUT with DiSPLAY Mode 1 → Disable clock signal</td><td>B1</td></tr><tr><td>Enable clock signal → Load temperature value from I2C Single Master Interface → Load LUT with DiSPLAY Mode 2 → Disable clock signal</td><td>B9</td></tr><tr><td>Enable clock signal → Enable Analog → Display with DiSPLAY Mode 1 → Disable Analog → Disable OSC</td><td>C7</td></tr><tr><td>Enable clock signal → Enable Analog → Display with DiSPLAY Mode 2 → Disable Analog → Disable OSC</td><td>CF</td></tr><tr><td>Enable clock signal → Enable Analog → Load temperature value from I2C Single Master Interface → Load temperature value from I2C Single Master Interface → DiSPLAY with DiSPLAY Mode 1 → Disable Analog → Disable OSC</td><td>F7</td></tr><tr><td>Enable clock signal → Enable Analog → Load temperature value from I2C Single Master Interface → Load temperature value from I2C Single Master Interface → DiSPLAY with DiSPLAY Mode 2 → Disable Analog → Disable OSC</td><td>FF</td></tr></table>		Operating sequence	Parameter (in Hex)	Enable clock signal	80	Disable clock signal	01	Enable clock signal → Enable Analog	C0	Disable Analog → Disable clock signal	03	Enable clock signal → Load LUT with DiSPLAY Mode 1 → Disable clock signal	91	Enable clock signal → Load LUT with DiSPLAY Mode 2 → Disable clock signal	99	Enable clock signal → Load temperature value from I2C Single Master Interface → Load LUT with DiSPLAY Mode 1 → Disable clock signal	B1	Enable clock signal → Load temperature value from I2C Single Master Interface → Load LUT with DiSPLAY Mode 2 → Disable clock signal	B9	Enable clock signal → Enable Analog → Display with DiSPLAY Mode 1 → Disable Analog → Disable OSC	C7	Enable clock signal → Enable Analog → Display with DiSPLAY Mode 2 → Disable Analog → Disable OSC	CF	Enable clock signal → Enable Analog → Load temperature value from I2C Single Master Interface → Load temperature value from I2C Single Master Interface → DiSPLAY with DiSPLAY Mode 1 → Disable Analog → Disable OSC	F7	Enable clock signal → Enable Analog → Load temperature value from I2C Single Master Interface → Load temperature value from I2C Single Master Interface → DiSPLAY with DiSPLAY Mode 2 → Disable Analog → Disable OSC	FF
Operating sequence	Parameter (in Hex)																																					
Enable clock signal	80																																					
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Enable clock signal → Load LUT with DiSPLAY Mode 1 → Disable clock signal	91																																					
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Enable clock signal → Load temperature value from I2C Single Master Interface → Load LUT with DiSPLAY Mode 1 → Disable clock signal	B1																																					
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Enable clock signal → Enable Analog → Load temperature value from I2C Single Master Interface → Load temperature value from I2C Single Master Interface → DiSPLAY with DiSPLAY Mode 1 → Disable Analog → Disable OSC	F7																																					
Enable clock signal → Enable Analog → Load temperature value from I2C Single Master Interface → Load temperature value from I2C Single Master Interface → DiSPLAY with DiSPLAY Mode 2 → Disable Analog → Disable OSC	FF																																					

Command Table												Command	Description
R/W	D/C#	Hex	D7	D6	D5	D4	D3	D2	D1	D0			
0	0	24	0	0	1	0	0	1	0	0		Write RAM (Black White) / RAM 0x24	After this command, data entries will be written into the BW RAM until another command is written. Address pointers will advance accordingly. For White pixel: Content of Write RAM(BW) = 1 For Black pixel: Content of Write RAM(BW) = 0
0	0	25	0	0	1	0	0	1	0	1		Write RAM (Dithering)	After this command, data entries will be written into the dithering engine.
0	0	26	0	0	1	0	0	1	1	0		Write RAM (RED) / RAM 0x26	After this command, data entries will be written into the RED RAM until another command is written. Address pointers will advance accordingly. For Red pixel: Content of Write RAM(RED) = 1 For non-Red pixel [Black or White]: Content of Write RAM(RED) = 0
0	0	27	0	0	1	0	0	1	1	1		Read RAM	After this command, data read on the MCU bus will fetch data from RAM [According to parameter of Register 41h to select reading RAM(BW) / RAM(RED)], until another command is written. Address pointers will advance accordingly. The 1st byte of data read is dummy data.
0	0	28	0	0	1	0	1	0	0	0		VCOM Sense	Enter VCOM sensing conditions and hold for duration defined in 29h before reading VCOM value. The sensed VCOM voltage is stored in register The command required CLKEN=1 and ANALOGEN=1 Refer to Register 0x22 for detail. BUSY pad will output high during operation.
0	0	29	0	0	1	0	1	0	0	1		VCOM Sense Duration	Stabling time between entering VCOM sensing mode and reading acquired. A[6]=1, Normal Mode A[6]=0, Reserve A[3:0] = 09h, duration = 10s. VCOM sense duration = Setting + 1 Seconds
0	1		0	A ₆	0	0	A ₃	A ₂	A ₁	A ₀			

Command Table												
R/W#	D/C#	Hex	D7	D6	D5	D4	D3	D2	D1	D0	Command	Description
0	0	2E	0	0	1	0	1	1	1	0	User ID Read	Read 10 Byte User ID stored in OTP: A[7:0]]~J[7:0]: UserID (R38, Byte A and Byte J) [10 bytes]
1	1		A7	A6	A5	A4	A3	A2	A1	A0		
1	1		B7	B6	B5	B4	B3	B2	B1	B0		
1	1		C7	C6	C5	C4	C3	C2	C1	C0		
1	1		D7	D6	D5	D4	D3	D2	D1	D0		
1	1		E7	E6	E5	E4	E3	E2	E1	E0		
1	1		F7	F6	F5	F4	F3	F2	F1	F0		
1	1		G7	G6	G5	G4	G3	G2	G1	G0		
1	1		H7	H6	H5	H4	H3	H2	H1	H0		
1	1		I7	I6	I5	I4	I3	I2	I1	I0		
1	1		J7	J6	J5	J4	J3	J2	J1	J0		
0	0	2F	0	0	1	0	1	1	1	1	Status Bit Read	Read IC status Bit [POR 0x01] A[5]: HV Ready Detection flag [POR=0] 0: Ready 1: Not Ready A[4]: VCI Detection flag [POR=0] 0: Normal 1: VCI lower than the Detect level A[3]: [POR=0] A[2]: Busy flag [POR=0] 0: Normal 1: BUSY A[1:0]: Chip ID [POR=01] Remark: A[5] and A[4] status are not valid after RESET, they need to be initiated by command 0x14 and command 0x15 respectively.
1	1		0	0	A5	A4	0	0	A1	A0		
0	0	30	0	0	1	1	0	0	0	0	Program WS OTP	Program OTP of Waveform Setting The contents should be written into RAM before sending this command. The command required CLKEN=1. Refer to Register 0x22 for detail. BUSY pad will output high during operation.
0	0	31	0	0	1	1	0	0	0	1	Load WS OTP	Load OTP of Waveform Setting The command required CLKEN=1. Refer to Register 0x22 for detail. BUSY pad will output high during operation.
0	0	32	0	0	1	1	0	0	1	0	Write LUT register	Write LUT register from MCU interface [105 bytes], which contains the content of VS [nX-LUT], TP #[nX], RP#[n].
0	1		A7	A6	A5	A4	A3	A2	A1	A0		
0	1		B7	B6	B5	B4	B3	B2	B1	B0		
0	1		:	:	:	:	:	:	:	:		
0	1		:	:	:	:	:	:	:	:		

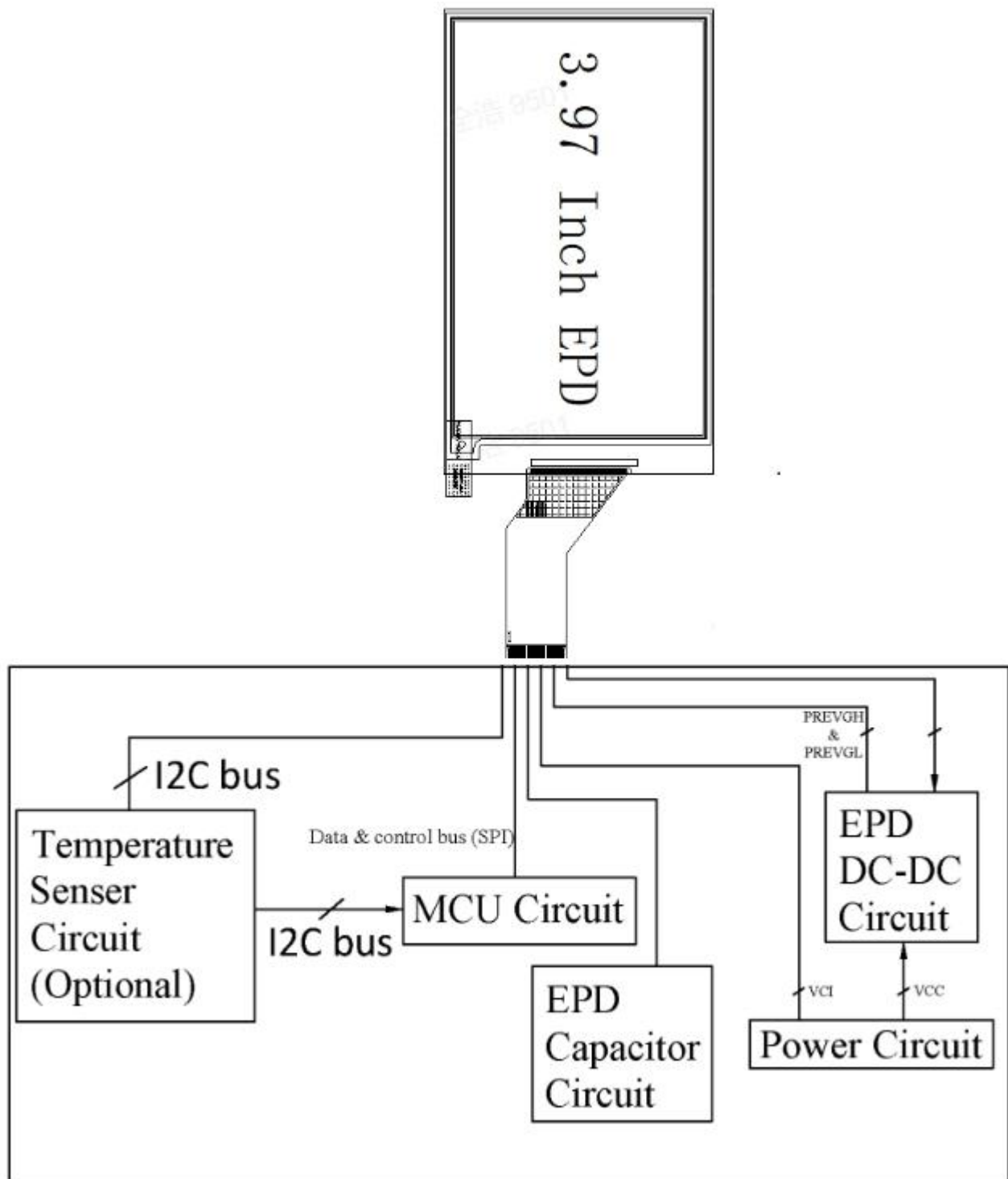
Command Table													Command	Description
R/W#	D/C#	Hex	D7	D6	D5	D4	D3	D2	D1	D0				
0	0	34	0	0	1	1	0	1	0	0	CRC calculation		CRC calculation command	BUSY pad will output high during operation.
0	0	35	0	0	1	1	0	1	0	1	CRC Status Read		CRC Status Read	A[15:0] is the CRC read out value
1	1		A ₁₅	A ₁₄	A ₁₃	A ₁₂	A ₁₁	A ₁₀	A ₉	A ₈				
1	1		A ₇	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀				
0	0	36	0	0	1	1	0	1	1	0	Program OTP selection		Program OTP Selection according to the OTP Selection Control [R37h and R38h]	The command required CLKEN=1. Refer to Register 0x22 for detail. BUSY pad will output high during operation.
0	0	37	0	0	1	1	0	1	1	1	Write Register for Display Option		Write Register for Display Option	
0	1		0	0	0	0	0	0	0	0			B[7:0] Display Mode for WS[7:0]	
0	1		B ₇	B ₆	B ₅	B ₄	B ₃	B ₂	B ₁	B ₀			C[7:0] Display Mode for WS[15:8]	
0	1		C ₇	C ₆	C ₅	C ₄	C ₃	C ₂	C ₁	C ₀			D[7:0] Display Mode for WS[23:16]	
0	1		D ₇	D ₆	D ₅	D ₄	D ₃	D ₂	D ₁	D ₀			E[7:0] Display Mode for WS[31:24]	
0	1		E ₇	E ₆	E ₅	E ₄	E ₃	E ₂	E ₁	E ₀			F[3:0] Display Mode for WS[35:32]	
0	1		0	F ₆	0	0	F ₃	F ₂	F ₁	F ₀			0: Display Mode 1	
0	1		G ₇	G ₆	G ₅	G ₄	G ₃	G ₂	G ₁	G ₀			1: Display Mode 2	
0	1		H ₇	H ₆	H ₅	H ₄	H ₃	H ₂	H ₁	H ₀			F[6]: RAM Ping-Pong for Display Mode 2	
0	1		I ₇	I ₆	I ₅	I ₄	I ₃	I ₂	I ₁	I ₀			1: RAM ping-pong enable	
0	1		J ₇	J ₆	J ₅	J ₄	J ₃	J ₂	J ₁	J ₀			0: RAM ping-pong disable	
													G[7:0]~J[7:0] module ID /waveform version.	
													Remarks:	
													1) A[7:0]~J[7:0] can be stored in OTP	
													2) RAM ping-pong function is not support for Display Mode 1	
0	0	38	0	0	1	1	1	0	0	0	Write Register for User ID		Write Register for User ID	
0	1		A ₇	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀			A[7:0]~J[7:0]: UserID [10 bytes]	
0	1		B ₇	B ₆	B ₅	B ₄	B ₃	B ₂	B ₁	B ₀				
0	1		C ₇	C ₆	C ₅	C ₄	C ₃	C ₂	C ₁	C ₀				
0	1		D ₇	D ₆	D ₅	D ₄	D ₃	D ₂	D ₁	D ₀				
0	1		E ₇	E ₆	E ₅	E ₄	E ₃	E ₂	E ₁	E ₀				
0	1		F ₇	F ₆	F ₅	F ₄	F ₃	F ₂	F ₁	F ₀				
0	1		G ₇	G ₆	G ₅	G ₄	G ₃	G ₂	G ₁	G ₀				
0	1		H ₇	H ₆	H ₅	H ₄	H ₃	H ₂	H ₁	H ₀				
0	1		I ₇	I ₆	I ₅	I ₄	I ₃	I ₂	I ₁	I ₀				
0	1		J ₇	J ₆	J ₅	J ₄	J ₃	J ₂	J ₁	J ₀				

Command Table																																										
R/W#	DirC#	Hex	D7	D6	D5	D4	D3	D2	D1	D0	Command	Description																														
0	0	39	0	0	1	1	1	0	0	1	OTP program mode	OTP program mode A[1:0] = 00: Normal Mode [POR] A[1:0] = 11: Internal generated OTP programming voltage Remark: User is required to EXACTLY follow the reference code sequences																														
0	1		0	0	0	0	0	0	A ₁	A ₀																																
0	0	3A	0	0	1	1	1	0	1	0	Reserved	Reserved																														
0	0	3B	0	0	1	1	1	0	1	1	Reserved	Reserved																														
0	0	3C	0	0	1	1	1	1	0	0	Border Waveform Control	Select border waveform for VBD A[7:0] = C0h [POR], set VBD as HIZ. A [7:6] :Select VBD option <table><tr><td>A[7:6]</td><td>Select VBD as</td></tr><tr><td>00</td><td>GS Transition, Defined in A[1:0]</td></tr><tr><td>01</td><td>Fix Level, Defined in A[5:4]</td></tr><tr><td>10</td><td>VCOM</td></tr><tr><td>11[POR]</td><td>HIZ</td></tr></table> A [5:4] Fix Level Setting for VBD <table><tr><td>A[5:4]</td><td>VBD level</td></tr><tr><td>00[POR]</td><td>VSS</td></tr><tr><td>01</td><td>VSH1</td></tr><tr><td>10</td><td>VSL</td></tr><tr><td>11</td><td>VSH2</td></tr></table> A [1:0] GS Transition setting for VBD <table><tr><td>A[1:0]</td><td>VBD Transition</td></tr><tr><td>00[POR]</td><td>LUT0</td></tr><tr><td>01</td><td>LUT1</td></tr><tr><td>10</td><td>LUT2</td></tr><tr><td>11</td><td>LUT3</td></tr></table>	A[7:6]	Select VBD as	00	GS Transition, Defined in A[1:0]	01	Fix Level, Defined in A[5:4]	10	VCOM	11[POR]	HIZ	A[5:4]	VBD level	00[POR]	VSS	01	VSH1	10	VSL	11	VSH2	A[1:0]	VBD Transition	00[POR]	LUT0	01	LUT1	10	LUT2	11	LUT3
A[7:6]	Select VBD as																																									
00	GS Transition, Defined in A[1:0]																																									
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11	LUT3																																									
0	1		A ₇	A ₆	A ₅	A ₄	0	0	A ₁	A ₀																																
0	0	41	0	1	0	0	0	0	0	1	Read RAM Option	Read RAM Option A[0]= 0 [POR] 0: Read RAM corresponding to RAM 0x24 1: Read RAM corresponding to RAM 0x26																														
0	1		0	0	0	0	0	0	0	A ₀																																

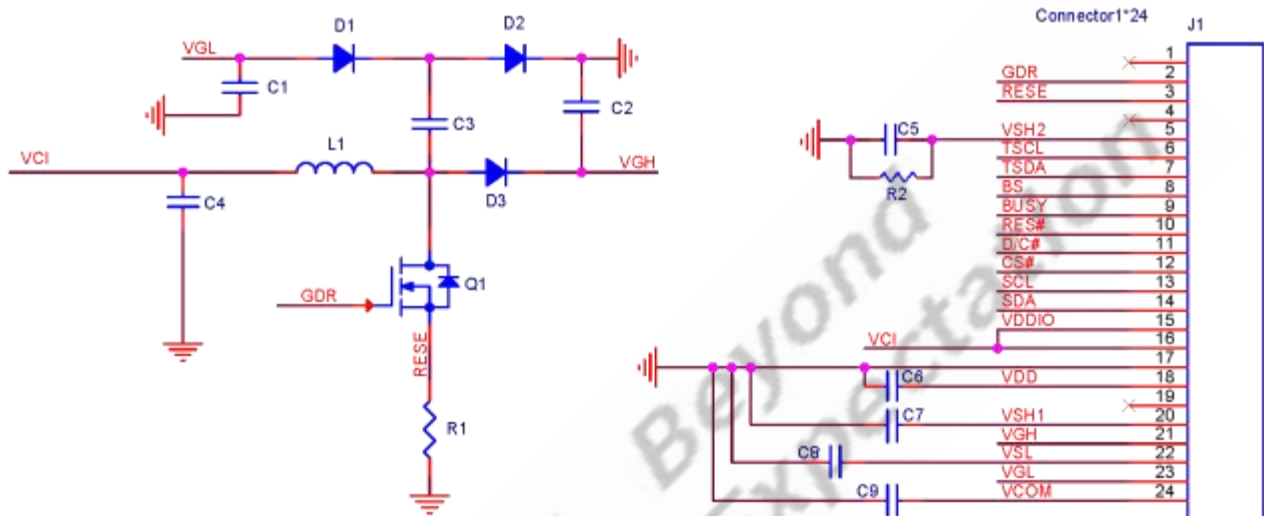
Command Table																																																				
R/W#	D/C#	Hex	D7	D6	D5	D4	D3	D2	D1	D0	Command	Description																																								
0	0	44	0	1	0	0	0	1	0	0	Set RAM X - address Start / End position	Specify the start/end positions of the window address in the X direction by an address unit for RAM A[9:0]: XSA[9:0], XStart, POR = 000h B[9:0]: XEA[9:0], XEnd, POR = 3BFh																																								
0	1		A ₇	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀																																										
0	1		-	-	-	-	-	-	A ₉	A ₈																																										
0	1		0	0	B ₅	B ₄	B ₃	B ₂	B ₁	B ₀																																										
0	1		-	-	-	-	-	-	B ₉	B ₈																																										
0	0	45	0	1	0	0	0	1	0	1	Set RAM Y- address Start / End position	Specify the start/end positions of the window address in the Y direction by an address unit for RAM A[9:0]: YSA[9:0], YStart, POR = 000h B[9:0]: YEA[9:0], YEnd, POR = 2A7h																																								
0	1		A ₇	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀																																										
0	1		-	-	-	-	-	-	A ₉	A ₈																																										
0	1		B ₇	B ₆	B ₅	B ₄	B ₃	B ₂	B ₁	B ₀																																										
0	1		-	-	-	-	-	-	B ₉	B ₈																																										
0	0	46	0	1	0	0	0	1	1	0	Auto Write RED RAM for Regular Pattern	Auto Write RED RAM for Regular Pattern A[7:0] = 00h [POR] A[7]: The 1st step value, POR = 0 A[6:4]: Step Height, POR= 000 Step of alter RAM in Y-direction according to Gate <table><tr><th>A[6:4]</th><th>Height</th><th>A[6:4]</th><th>Height</th></tr><tr><td>000</td><td>8</td><td>100</td><td>128</td></tr><tr><td>001</td><td>16</td><td>101</td><td>256</td></tr><tr><td>010</td><td>32</td><td>110</td><td>512</td></tr><tr><td>011</td><td>64</td><td>111</td><td>680</td></tr></table> A[2:0]: Step Width, POR= 000 Step of alter RAM in X-direction according to Source <table><tr><th>A[2:0]</th><th>Width</th><th>A[2:0]</th><th>Width</th></tr><tr><td>000</td><td>8</td><td>100</td><td>128</td></tr><tr><td>001</td><td>16</td><td>101</td><td>256</td></tr><tr><td>010</td><td>32</td><td>110</td><td>512</td></tr><tr><td>011</td><td>64</td><td>111</td><td>960</td></tr></table> BUSY pad will output high during operation.	A[6:4]	Height	A[6:4]	Height	000	8	100	128	001	16	101	256	010	32	110	512	011	64	111	680	A[2:0]	Width	A[2:0]	Width	000	8	100	128	001	16	101	256	010	32	110	512	011	64	111	960
A[6:4]	Height	A[6:4]	Height																																																	
000	8	100	128																																																	
001	16	101	256																																																	
010	32	110	512																																																	
011	64	111	680																																																	
A[2:0]	Width	A[2:0]	Width																																																	
000	8	100	128																																																	
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011	64	111	960																																																	
0	1		A ₇	A ₆	A ₅	A ₄	0	A ₂	A ₁	A ₀																																										

Command Table												Command	Description																																								
R/W#	D/C#	Hex	D7	D6	D5	D4	D3	D2	D1	D0																																											
0	0	47	0	1	0	0	0	1	1	1	Auto Write B/W RAM for Regular Pattern	Auto Write B/W RAM for Regular Pattern A[7:0] = 00h [POR] A[7]: The 1st step value, POR = 0 A[6:4]: Step Height, POR= 000 Step of alter RAM in Y-direction according to Gate <table><tr><th>A[6:4]</th><th>Height</th><th>A[6:4]</th><th>Height</th></tr><tr><td>000</td><td>8</td><td>100</td><td>128</td></tr><tr><td>001</td><td>16</td><td>101</td><td>256</td></tr><tr><td>010</td><td>32</td><td>110</td><td>512</td></tr><tr><td>011</td><td>64</td><td>111</td><td>680</td></tr></table> A[2:0]: Step Width, POR= 000 Step of alter RAM in X-direction according to Source <table><tr><th>A[2:0]</th><th>Width</th><th>A[2:0]</th><th>Width</th></tr><tr><td>000</td><td>8</td><td>100</td><td>128</td></tr><tr><td>001</td><td>16</td><td>101</td><td>256</td></tr><tr><td>010</td><td>32</td><td>110</td><td>512</td></tr><tr><td>011</td><td>64</td><td>111</td><td>960</td></tr></table> During operation, BUSY pad will output high.	A[6:4]	Height	A[6:4]	Height	000	8	100	128	001	16	101	256	010	32	110	512	011	64	111	680	A[2:0]	Width	A[2:0]	Width	000	8	100	128	001	16	101	256	010	32	110	512	011	64	111	960	
A[6:4]	Height	A[6:4]	Height																																																		
000	8	100	128																																																		
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0	1		A7	A6	A5	A4	0	A2	A1	A0																																											
0	0	4D	0	1	0	0	1	1	0	1	Dithering engine Start/Stop	This command is used for start and stop dithering function, start commands should be sent before Command 0x25 and stop commands should be sent after the data bytes of command 0x25 <table><tr><th></th><th colspan="4">Data byte</th></tr><tr><th></th><th>A[7:0]</th><th>B[7:0]</th><th>C[7:0]</th><th>D[7:0]</th></tr><tr><td>Start Sierra Lite</td><td>80</td><td>00</td><td>78</td><td>00</td></tr><tr><td>Start Floyd-Steinberg</td><td>83</td><td>00</td><td>78</td><td>00</td></tr><tr><td>Stop dithering</td><td>00</td><td>00</td><td>78</td><td>00</td></tr></table>		Data byte					A[7:0]	B[7:0]	C[7:0]	D[7:0]	Start Sierra Lite	80	00	78	00	Start Floyd-Steinberg	83	00	78	00	Stop dithering	00	00	78	00																
	Data byte																																																				
	A[7:0]	B[7:0]	C[7:0]	D[7:0]																																																	
Start Sierra Lite	80	00	78	00																																																	
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Stop dithering	00	00	78	00																																																	
0	1		A7	A6	A5	A4	A3	A2	A1	A0																																											
0	1		B7	B6	B5	B4	B3	B2	B1	B0																																											
0	1		C7	C6	C5	C4	C3	C2	C1	C0																																											
0	1		D7	D6	D5	D4	D3	D2	D1	D0																																											
0	0	4E	0	1	0	0	1	1	1	0	Set RAM X address counter	Make initial settings for the RAM X address in the address counter (AC) A[9:0]: 000h [POR].																																									
0	1		A7	A6	A5	A4	A3	A2	A1	A0																																											
0	1		0	0	0	0	0	0	0	A8				A8																																							
0	0	4F	0	1	0	0	1	1	1	1	Set RAM Y address counter	Make initial settings for the RAM Y address in the address counter (AC) A[9:0]: 000h [POR].																																									
0	1		A7	A6	A5	A4	A3	A2	A1	A0																																											
0	1		0	0	0	0	0	0	0	A8				A8																																							
0	0	7F	0	1	1	1	1	1	1	1	NOP	This command is an empty command; it does not have any effect on the display module. However, it can be used to terminate Frame Memory Write or Read Commands.																																									

8. Block Diagram



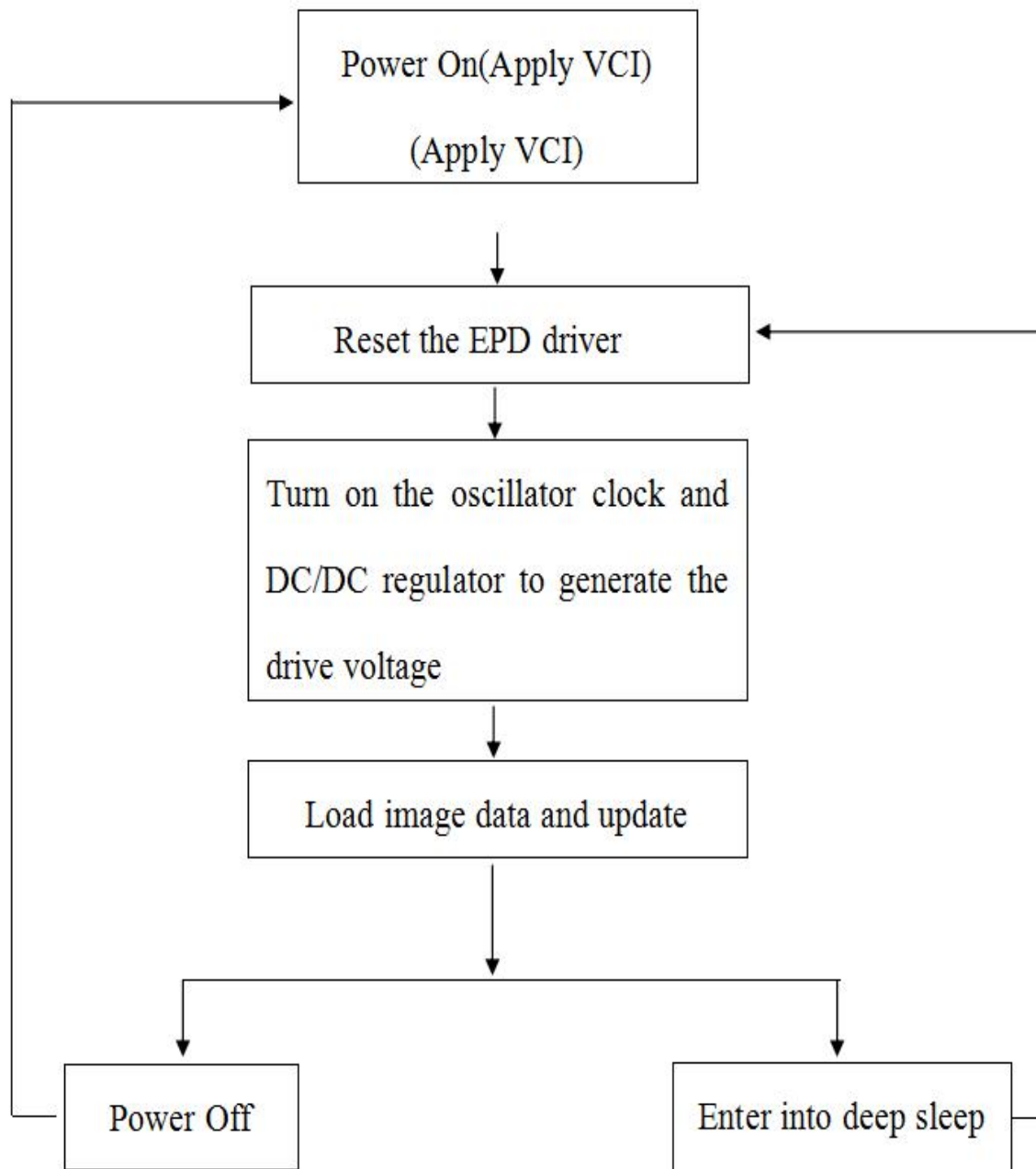
9. Typical Application Circuit with SPI Interface



Part Name	Value	Reference Part /Requirements for spare part
C4 C6	1uF	0603;X5R/X7R;Voltage Rating:6v
C1 C2 C3 C5 C7 C8	4.7uF	0805; X5R/X7R;Voltage Rating:25v
C9	1uF	0805; X7R;Voltage Rating:25v
R1	2.2Ohm	0805; 1% variation,
R2	100KOhm	0603/0805; 1% variation,
D1 D2 D3	Diode	MBR0530
Q1	NMOS	Si1304BDL
L1	47UH	CDRH2D18/LDNP-470NC

10 Typical Operating Sequence

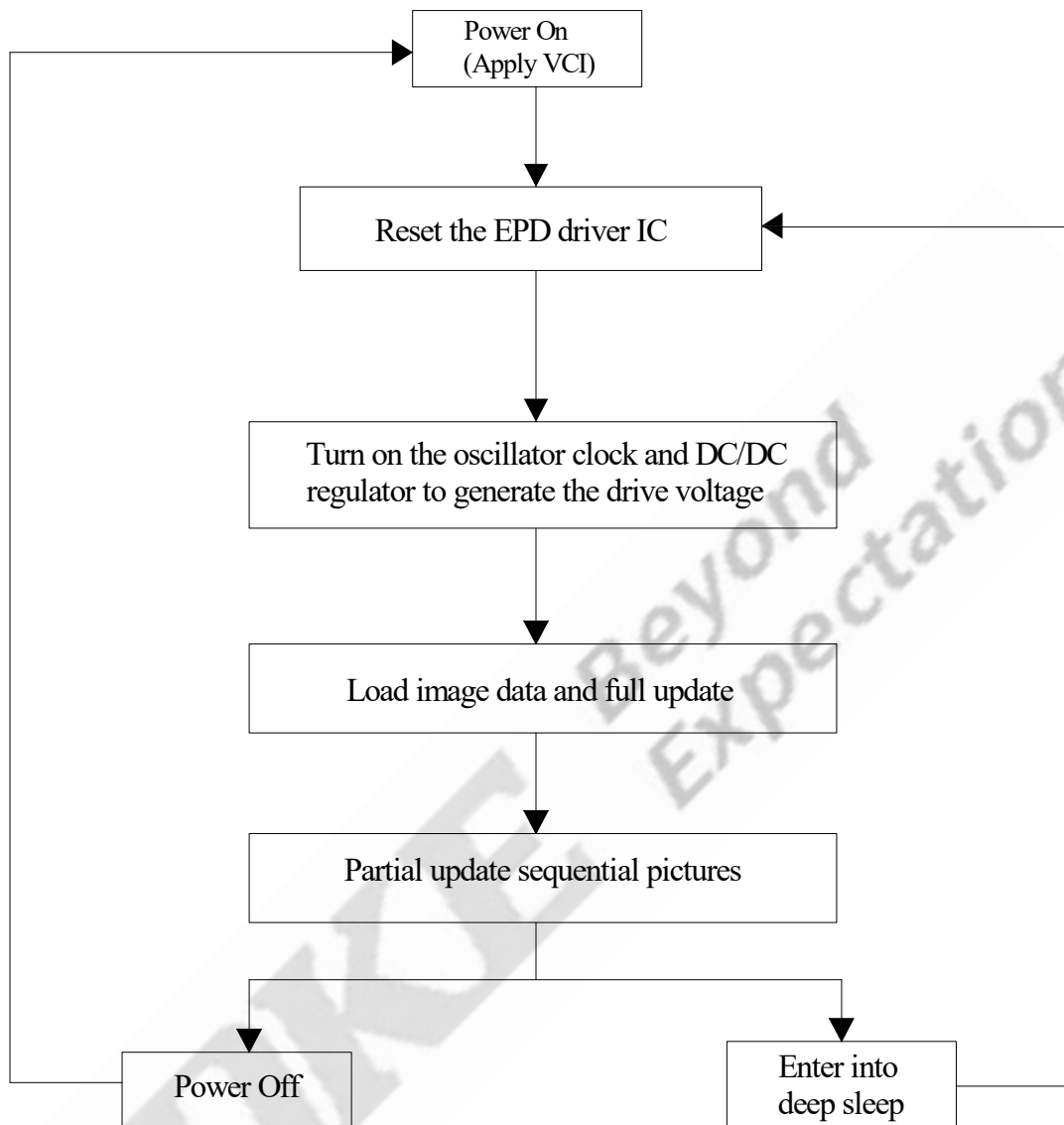
10.1 OTP Operation Flow



10.2 OTP Operation Reference Program Code

ACTION	VALUE/DATA	COMMENT
POWER ON		
delay	10ms	
PIN CONFIG		
RES#	low	Hardware reset
delay	200us	
RES#	high	
delay	200us	
Read busy pin		Wait for busy low
Command 0x12		Software reset
Read busy pin		Wait for busy low
SET VOLTAGE AND LOAD LUT		
LOAD IMAGE AND UPDATE		
Command 0x24	48000bytes	Load image (800/8*480)(BW)
Command 0x20		
Read busy pin		Wait for busy low
Command 0x10	Data 0X01	Enter deep sleep mode
POWER OFF		

10.3 OTP Partial update Operation Flow



11. Reliability Test

NO	Test items	Test condition
1	Low-Temperature Storage	T = -25°C, 240 h Test in white pattern
2	High-Temperature, Low-Humidity Storage	T=+60°C, RH=35%, 168h Test in white pattern
3	High-Temperature, Low-Humidity Operation	T=40°C, RH=35%, 240h
4	Low-Temperature Operation	0°C, 240h
5	High-Temperature, High-Humidity Operation	T=40°C, RH=80%, 240h
6	High Temperature, High Humidity Storage	T=50°C, RH=80%, 240h Test in white pattern
7	Temperature Cycle	1 cycle:[-25°C 30min]→[+60 °C 30 min] : 100 cycles Test in white pattern
8	ESD 静电	接触放电+/-2KV(HBM C:100pF;R:1.5k Ω) 接触放电+/-200V(MM C:200pF;R:0Ω) (裸屏 EPD 显示, 包括 IC 和 FPC 区域)

Note: 1. Stay white pattern for storage and non-operation test.
2. Operation is black→white pattern, the interval is 150s.

12. Quality Assurance

12.1 Environment

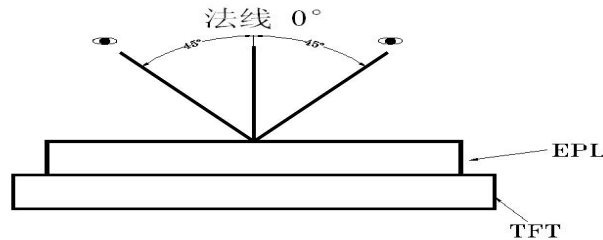
Temperature: $23 \pm 3^{\circ}\text{C}$

Humidity: $55 \pm 10\% \text{RH}$

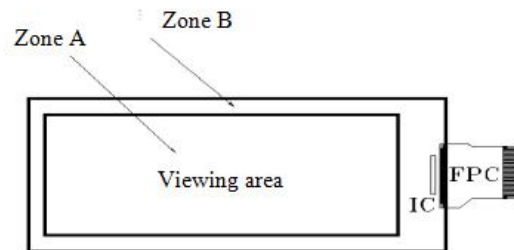
12.2 Illuminance

Brightness: 1200~1500LUX; distance: 20~30CM; Angle: Relate 45° surround.

12.3 Inspect method

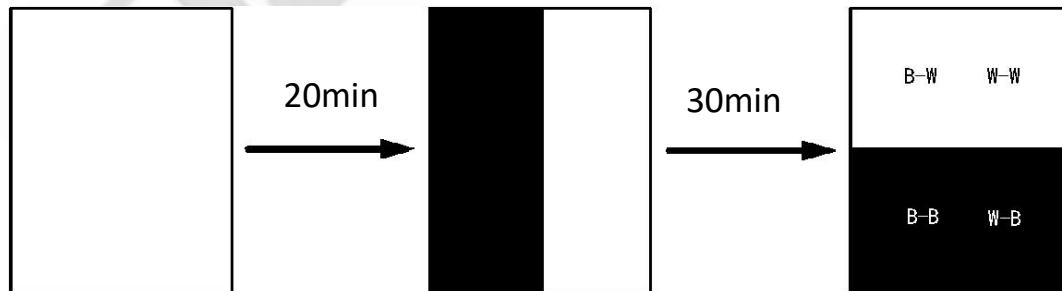


12.4 Display area



12.5 Ghosting test method

Two-color ghosting is measured with following transition from horizontal 2 scale pattern to vertical 2 scale pattern. The listed optical characteristics are only guaranteed under the controller & waveform provided by DKE.



1) Measurement Instruments: X-rite i1Pro

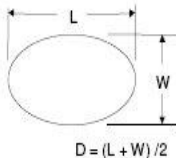
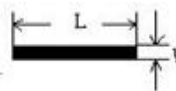
2) Ghosting formula:

W ghosting: $\Delta L = \text{Max} (\Delta L(\text{W-W}, \text{B-W})) - \text{Min} (\Delta L(\text{W-W}, \text{B-W}))$

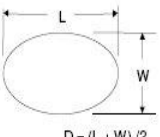
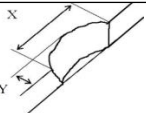
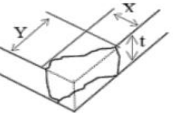
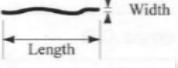
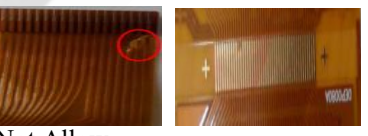
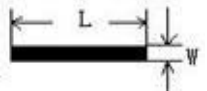
K ghosting: $\Delta L = \text{Max} (\Delta L(\text{W-B}, \text{B-B})) - \text{Min} (\Delta L(\text{W-B}, \text{B-B}))$

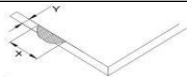
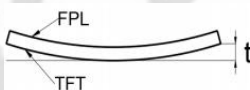
12.6 Inspection standard

12.6.1 Electric inspection standard

NO.	Item	Standard	Defect level	Method	Scope
1	Display	Clear display、 Display complete、 Display uniform	MA	Visual inspection	Zone A
2	Black/White spots	 $D = (L + W) / 2$ $D \leq 0.3\text{mm}$, negligible $0.3\text{mm} < D \leq 0.5\text{mm}$, $N \leq 5$, Allowed $0.5\text{mm} < D$ Not Allow	MI	Visual Inspection card	Zone A
3	Black/White lines (No switch)	 $L \leq 1.0\text{mm}$, $W \leq 0.15\text{mm}$ negligible $1.0\text{mm} < L \leq 4.0\text{mm}$ $0.15\text{mm} < W \leq 0.5\text{mm}$ $N \leq 4$ allowable $L > 4.0\text{mm}$,or $W > 0.5\text{mm}$ is not allowed	MI	Visual Inspection card	Zone A
4	Ghost image	Allowed in switching process	MI	Visual	Zone A
5	Flash dot / Multilateral	Flash points are allowed when switching screens Multilateral colors outside the frame are allowed for fixed screen time	MI	Visual/ Inspection card	Zone A
6	Segmented display	Selection segments are all displayed, and other segments are not displayed after the selection segment.	MA	Visual inspection card	Zone A
7	Short circuit/ Circuit break/ Abnormal Display	Not Allow	MA	Visual	Zone A
8	Corner mura	$X > 1\text{mm}$, $Y > 1\text{mm}$, not allow 	MI	Visual/Ruler	Zone A

12.6.2 Appearance inspection standard

NO.	Item	Standard	Defect level	Method	Scope
1	B/W spots /Bubble/ Foreign bodies/ Dents	 $D = (L + W) / 2$ $D \leq 0.3\text{mm}$, Allowed $0.3\text{mm} < D \leq 0.5\text{mm}$, $N \leq 5$ $D > 0.5\text{mm}$, Not Allow	MI	Visual/ inspection Card	Zone A
2	Chips/Scratch/ Edge crown	 $X \leq 3\text{mm}$, $Y \leq 0.5\text{mm}$ $t =$ not counted, and without affecting the electrode , permissible  $X \leq 2\text{mm}$ or $Y \leq 2\text{mm}$ $t =$ not counted.and without affecting the electrode , permissible  $W \leq 0.1\text{mm}$, $L \leq 5\text{mm}$, without affecting the electrode , $n \leq 2$	MI	Visual / Ruler	Zone A/B
3	TFT Cracks	 Not Allow	MA	Visual / Microscope	Zone A/B
4	Dirty/ foreign body	Allowed if can be removed	MI	Visual / clean cloth	Zone A / B
5	FPC broken/ FPC oxidation / scratch	 Not Allow	MA	Visual / Microscope	Zone B
6	B/W Line	 $L \leq 1.0\text{mm}$, $W \leq 0.15\text{mm}$ negligible $1.0\text{mm} < L \leq 4.0\text{mm}$ $0.15\text{mm} < W \leq 0.5\text{mm}$ $N \leq 4$ allowable $L > 4.0\text{mm}$, $W > 0.5\text{mm}$ is not allowed	MI	Visual / Ruler	Zone A

7	TFT edge bulge /TFT chromatic aberration	 TFT edge bulge: $X \leq 3\text{mm}$, $Y \leq 0.3\text{mm}$ Allowed TFT chromatic aberration :Allowed	MI	Visual / Microscope	Zone A/B
8	Electrostatic point	$D \leq 0.35\text{mm}$, allow $0.35\text{mm} < D \leq 0.55\text{mm}$, $n \leq 5$ allow $D > 0.55\text{mm}$ is not allowed	MI	Visual /inspection card	Zone A
9	PCB damaged/ Poor welding/ Curl	PCB (Circuit area) damaged Not Allow PCB Poor welding Not Allow PCB Curl $\leq 1\%$	MA	Visual / Ruler	Zone B
10	Edge glue height/ Edge glue bubble	Edge Adhesives $H \leq$ PS surface (Including protect film) Edge adhesives overflow onto protect film $\leq 1/2$ FPL to PS Margin width Length excluding Edge adhesives seep in $\leq 1/2$ FPL Margin width Length excluding Edge adhesives bubble: bubble Width $\leq 1/2$ Margin width; Length $\leq 5.0\text{mm}$. $n \leq 5$	MI	Visual / Ruler	Zone B
11	Protect film	Surface scratch but not effect protect function, Allow	MI	Visual	Zone A/B
12	Silicon glue	Thickness $\leq$ PS surface(With protect film): Full cover the IC; Shape: The width on the FPC $\leq 1.0\text{mm}$ (Front) The width on the FPC $\leq 1.0\text{mm}$ (Back) smooth surface, No obvious raised.	MI	Visual/Ruler	Zone B
13	Warp degree (TFT substrate)	 $t \leq 1.5\text{mm}$	MI	Ruler	Zone B
14	Color difference in COM area (Silver point area)	Allowed	MI	Visual	Zone B
15	Corner and edge damage of FPL	Corner and edge damage $\leq 1/2$ border width 	MI	Visual / Ruler	Zone A

13.Packaging



PACKING INSTRUCTION

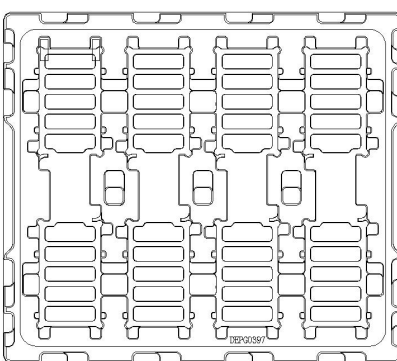
P/N	Customer Code	Ref. P/N	Type	PKG Method	Marking	Surface Marks	Pull Tape
DEPG0397			GLASS	Blister	BACK	None	YES

Packing Materials List					8PCS/LAYER, 20LAYER/CTN, TOTAL 160PCS/CTN.
List	Model	Materials	Q'ty	Unit	Pull tape: 
Carton	7# 417*362*229 mm	corrugate	1	Piece	
Inner Carton	7# (INNER) 400*343 *95 mm	corrugate	2	Piece	
Blister	DEPG03970	PET	22	Piece	
Thin foam	297*251 T1.5-1.8MM	EPE	20	Piece	
Antistatic vacuum bag	450*590*0.075		2	Piece	
Foam board	DKE2251-10	EPE	3	Piece	
PULL TAPE	16*5*T0.05		160	Piece	

Detail:

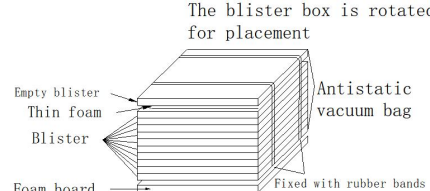
Blister box:

Note: there are 20 layers of products, divided into 2 inner boxes, and an empty blister box is placed on the top of each inner box, so the number of blister boxes is 22



QUANTITY: 8PCS

The blister box is rotated for placement



Empty blister
Thin foam
Blister
Foam board

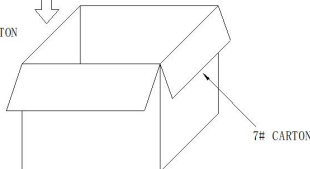
Fixed with rubber bands

PUT IT INTO 7# INNER CARTON

INNER BOX LABEL

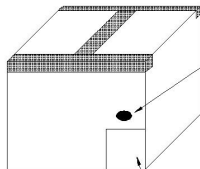
型号 (MODEL)	
数量 (QUANTITY)	
批号 (LOT#)	

PUT TWO 7# INNER CARTON INTO 7# CARTON



7# INNER CARTON
7# CARTON

Packing belt



rohs Label



Shipping marks according to customer's requirements

Epaper Identification	
QC:	PASS
Model No.	
Quantity.	pcs
Date:	
Carton No.	of

14. Handling, Safety, and Environment Requirements

Warning

The display glass may break when it is dropped or bumped on a hard surface. Handle with care. Should the display break, do not touch the electrophoretic material. In case of contact with electrophoretic material, wash with water and soap.

Caution

The display module should not be exposed to harmful gases, such as acid and alkali gases, which corrode electronic components. Disassembling the display module.

Disassembling the display module can cause permanent damage and invalidates the warranty agreements.

Observe general precautions that are common to handling delicate electronic components. The glass can break and front surfaces can easily be damaged. Moreover the display is sensitive to static electricity and other rough environmental conditions.

Data sheet status	
Product specification	This data sheet contains final product specifications.
Limiting values	
Limiting values given are in accordance with the Absolute Maximum Rating System (IEC 134). Stress above one or more of the limiting values may cause permanent damage to the device. These are stress ratings only and operation of the device at these or at any other conditions above those given in the Characteristics sections of the specification is not implied. Exposure to limiting values for extended periods may affect device reliability.	
Application information	
Where application information is given, it is advisory and does not form part of the specification.	
Product Environmental certification	
ROHS	
REMARK	
All The specifications listed in this document are guaranteed for module only. Post-assembled operation or component(s) may impact module performance or cause unexpected effect or damage and therefore listed specifications is not warranted after any Post-assembled operation.	
Transport environment	
When the humidity of transportation environment is between 45%RH~70%RH, the product can be stored for 30 days, and the product can be stored for 10 days if it is lower or higher than this range.	